# Mental Health in the Tech Industry

## CS 210:Data Management for Data Science


## Vanij Patel and Sean Lumasag


## Link: [https://github.com/seanlumasag/cs210-project](https://github.com/seanlumasag/cs210-project)
## Demo Video: video_project.mp4

**Research Question**: What workplace and personal factors predict whether a worker in the tech industry will seek mental health treatment, and how does mental health interfere with their work?

## 1. Problem Definition and Relevance

Mental health has become one of the most serious and underaddressed issues in the technical industry. Tech jobs are associated with high salaries and strong career paths and opportunities, but they also come with long hours, high pressure, and a culture where stress is seen as normal. This project looks at this issue directly and tries to predict whether a tech worker ends up seeking mental health treatment and to what extent mental health causes interference with work. The reason behind choosing this topic is to highlight the need to address this major issue in the tech world. Being students ourselves, aiming to break into these career paths, we are trying to showcase issues that should be addressed.

To support our question, we used the Open Sourcing Mental Illness survey datasets from 2014 and 2016. We evaluated all OSMI surveys from 2014 to 2021, but the chosen years were the only surveys that provided us with clean, structured datasets with the variables needed. Later OSMI surveys were harder to use because they had fewer usable responses, different schemas, or did not have the necessary fields for our project.

Prior research has looked at workplace mental health in general. Dewa et al. (2014) found that burnout is associated with decreased productivity and increased early age retirement, which highlights the real workplace costs of untreated mental conditions. In software engineering, Russo & Stol (2022) studied gender differences in personality traits amongst software engineers but did not examine treatment-seeking behavior or employer support. The Open Sourcing Mental Illness Survey has also appeared in prior Kaggle projects and in a merged dataset by Memon (2023), combining surveys from 2017–2021. However, that dataset only uses about 8–10 features and focuses on predicting current disorder status. No public analysis merges the 2014 and 2016 surveys with richer features, predicts treatment-seeking and work interference, or compares the predictive impact of personal factors versus workplace factors using separate models.

Our project makes various contributions, which set it apart from prior work. First, we built a 2,692-row unified dataset by manually schema-mapping two surveys that share no matching column names, though they represent the same data. Second, we predict treatment-seeking behavior rather than the disorder status, which better captures the decision to seek help. Third, we directly compare workplace and personal feature groups by training separate Random Forest models on each and comparing their Area Under Curve (AUC) score. Fourth, by merging the data, we add another column with the year, which allows us to compare treatment rate, stigma, and employer support and measure whether the year itself is a predictive feature. The reason behind this is that we acknowledge the years we are using are not recent, so we want to understand how big an impact it would make on the research question.

This project applies to concepts covered in CS 210 including database design, SQL querying for data analysis, data cleaning and normalization and feature engineering. These concepts were applied to various stages of our pipeline to help strengthen our end result.

## 2. Data and Methodology

### 2.1 Understanding Datasets

| Dataset | Source | Year | Rows | Columns |
|---|---|---|---|---|
| OSMI 2014 Survey | OSMI from Kaggle | 2014 | 1,259 | 27 |
| OSMI 2016 Survey | OSMI from Kaggle | 2016 | 1,433 | 63 |
| Merged | OSMI from Kaggle | 2014 + 2016 | 2,692 | Shared 26 |

The 2014 dataset uses short column identifiers like self_employed, work_interfere, and no_employees. The 2016 dataset uses longer question text as column names, such as "Are you self-employed?" This structural difference between the two datasets introduces the challenge of data engineering and schema mapping in this project.

**2.2 Data Pipeline**

Stage 1 - Analyzing Data

Both CSV files were loaded into the Pandas Dataframes and stored in a SQLite database for future querying. Before any cleaning, we ran null rate checks on both datasets. In 2014, three columns had over 20% null rates, which were comments(87%), state(40.9%), and work_interfere(21%). The comments column was almost empty and would not serve as useful in the analysis. The work_interfere null rate was important because that column turned out to be one of the most important predictors in our model, so handling its null values was crucial. The 2016 dataset had far more null columns, with 18. Nine of those exceeded 80%, though most of these columns existed only in the 2016 dataset and had too little data to be useful.

Stage 2 - Schema Alignment

This was the most intensive step, as the rest of the models were based on this. The 2016 survey has 63 columns written as full questions, and none match the 27 short column names in 2014. We built a 27-column mapping dictionary that translated each 2016 question to its 2014 equivalent. An example is "Are you self-employed?" to self_employed. The mapping was done manually by reading the 2016 questions and matching them conceptually to the 2014 column. 36 columns from 2016 had no equivalent in 2014, and therefore were left unmapped. A survey_year column was added to both datasets before the merge to track which year each row came from.

Stage 3 - Data Cleaning

Five cleaning steps were applied, with each targeting a data quality issue:

1. Age: The raw columns contained invalid entries, which included negative numbers, unrealistic values above 100, and possible entries of the birth year. We converted age to numeric, set any value outside of 18 to 75 to NaN, and replaced those nulls with the median. The reason behind choosing this range was to reflect a realistic working population.

2. Gender: The gender column had several unique entries. People wrote things like 'boy', 'maile', 'femail', and 'femake'. We wrote a standardize_gender function that first checked for exact common matches like “male” or “f”, and then searched for substring matches against terms listed for Female, Male, and Non-binary. Anything that did not match went to Other. In 2014, this resulted in 994 Male, 251 Female, 9 Non-binary, and 5 Other. In 2016, it resulted in 1,060 Male, 348 Female, 18 Non-binary, and 7 Other.
3. Treatment: The 2014 dataset stored treatment as “Yes” or “No” strings. The 2016 dataset was already encoded as 0 or 1. We converted 2014 to binary integers, so both datasets were consistent before the merge.
4. self_employed and remote_work normalization: We realized that 2014 stored self_employed as 0 or 1 integers while 2016 stored it as “Yes” or “No” strings. Similarly, remote_work was “Yes” or “No” in 2014 but Always"/"Sometimes"/"Never" in 2016. If not fixed, get_dummies would have created separate features with meaningless duplication. We wrote normalize_yes_no and normalize_remote_work functions to standardize both columns before merging.
5. work_interfere: This was one of the more complex cleaning decisions we made. In 2014, work_interfere is a single column with 264 nulls. These nulls represented people for whom mental health does not apply to work, so we filled them with “Not Applicable” rather than dropping and losing data. In 2016, the survey split this into two separate questions, which were one for when treated effectively and one for when not treated effectively. We merged these back into a single column, which made the 2016 column directly comparable to 2014's single measure.
6. High-null column: We dropped any 2016 column with over 80% nulls. This removed 3 columns, which were the tech/IT role column (81.6%), client impact after disclosure (90%), and work time percentage affected (85.8%).

Stage 4 - Merging and Validation
After alignment and cleaning, we selected the 26 columns present in both datasets and concatenated them into a single DataFrame. The merged dataset has 2,692 rows, where 1,259 were from 2014, and 1,433 were from 2016. The treatment classes were 1,476 seekers versus 1,216 non-seekers, resulting in a rough 55/45 split that, though an imbalance, is manageable without resampling. After merging, we verified the remaining null counts. The 287 nulls across most workplace columns all come from the same 287 self-employed 2016 respondents. The 1,259 nulls in share_friends_family come entirely from the 2014 dataset, which did not include that question.

Stage 5 - Feature Engineering
We built two composite scores to capture the higher-level understanding that the individual columns did not represent.

Employer Support Score: We took six employer-related columns, which are benefits, care_options, wellness_program, seek_help, anonymity, leave, and scored each response on a 0 to 1 scale. "Yes" mapped to 1.0, "No" to 0.0, and uncertain responses like "Don't know" to 0.5. For the leave column, which had more responses, we used the following scale: "Very easy" = 1.0, "Somewhat easy" = 0.75, "Neither easy nor difficult" = 0.5, "Somewhat difficult" = 0.25, "Very difficult" = 0.0. We then averaged the six scores into a single employer_support_score. The mean across the merged dataset was 0.473, meaning that employers scored below the neutral 0.5 on average.

Workplace Stigma Score: We used four columns for this score. mental_health_consequence and obs_consequence directly measure stigma and were scored with "Yes" = 1.0 and "No" = 0.0. The coworkers and supervisor columns measure comfort in discussing mental health, so they were inverted before averaging, where a "Yes" becomes 0.0 on the stigma scale. The four scores were averaged into stigma_score. The mean was 0.379.

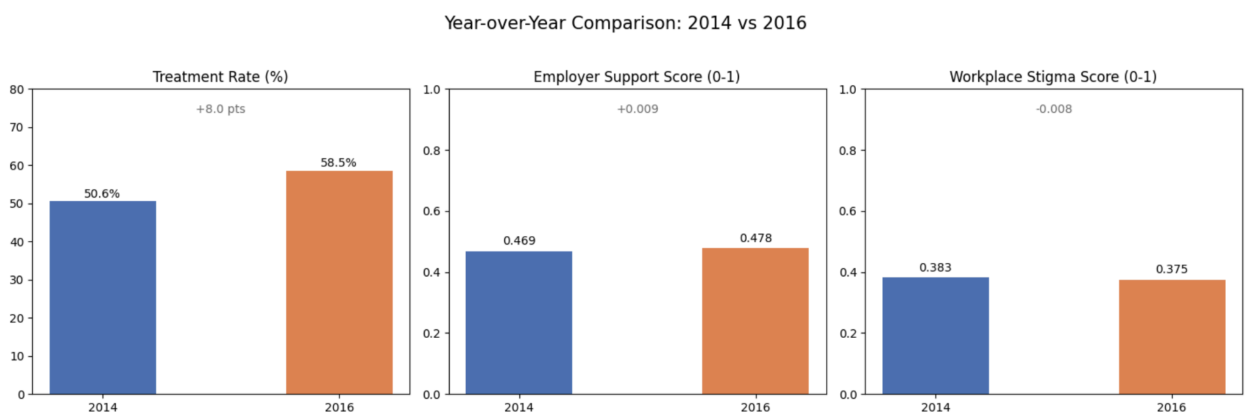
## 3. SQL Database Queries

We stored the merged dataset in SQLite and ran four analytical queries before modeling. These queries were chosen to identify patterns related to our research question.

**Q1 - Treatment Rate by Year**

| Year | Treatment Rate | Respondents |
|---|---|---|
| 2014 | 50.6 % | 1,259 |
| 2016 | 58.5% | 1,433 |

Treatment seeking increased by 7.9 percent between the two years. This is a meaningful shift for a two-year gap and likely reflects a growing awareness of mental health in the tech industry. This finding connects to our sub-experiment 3, where survey_year is ranked as 8th in feature importance, confirming that the year carries predictive power for other variables.

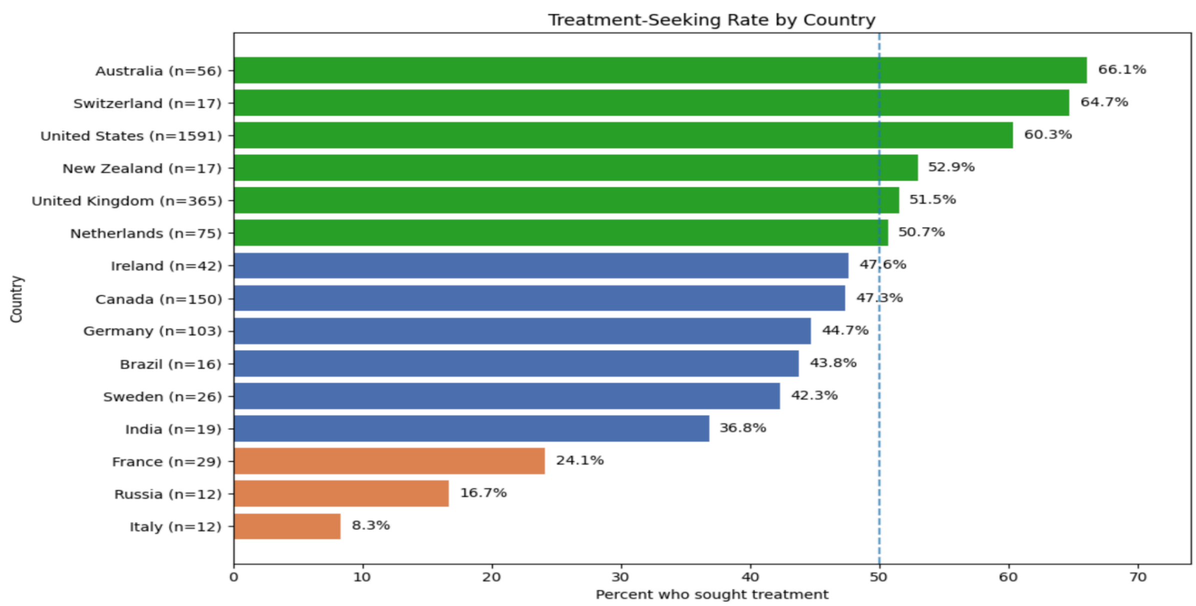



**Q2 - Treatment Rate by Company Size**

| Company Size | Treatment Rate | Respondents |
|---|---|---|
| 100-500 | 57.3% | 424 |
| More than 1000 | 56.3% | 538 |
| 1-5 | 55.9% | 222 |
| 26-100 | 53.9% | 581 |
| 500-1000 | 52.1% | 140 |
| 6-25 | 47.6% | 500 |

Company size shows a non-linear pattern as mid to large companies (100 - 500) have the highest treatment rates. A possible reason is that these companies may be large enough to offer mental health benefits while still being small enough for employees to have easy access. Very small companies (1 - 5) also show high rates, possibly because informal environments make it easier to have small conversations, as well as in front of a smaller team where you are closer. Small to mid-sized companies (6 - 25 and 26 - 100)  had the lowest rate, which may reflect a gap where companies are too small for formal programs.

**Q3 - Treatment Rate by Top 15 Countries (Table shows 5 below for spacing purposes)**
English-speaking countries show significantly higher treatment rates than European countries, but results should be interpreted carefully because of uneven sample sizes. This likely reflects a combination of healthcare access and cultural views towards mental health. It is also important to note that the survey is in English, so that could be a reason for the respondent counts. The dataset is heavily US-dominated, which is a limitation we later discuss. Though some countries in our top 15, such as Switzerland, have a really high percentage at 64.7%, the small count of 17 respondents makes that number less significant.




**Q4 - Treatment Rate by Work Interference Level**

| Work Interference | Treatment Rate | Count |
|---|---|---|
| Often | 88.1% | 691 |
| Sometimes | 73.8% | 836 |
| Rarely | 67.6% | 225 |
| Never | 15.6% | 224 |
| Not Applicable | 8.8% | 716 |

This is one of the more important findings, as it helps us conclude that as interference severity increases, treatment-seeking increases with it. Workers who say mental health often interferes with their work see a treatment rate of 88.1%, while those who say it never interferes is at 15.6%. This pattern tells us that treatment seeking is reactive, meaning people seek treatment when the problem is significant. This finding from Q4 and the feature importance ranking in the model reinforce each other.

## 4. Machine Learning Results

4.1 Model Setup

We trained two classifiers on the encoded dataset. Logistic Regression was used as a baseline model for interpretation, and Random Forest was used as the primary model for its feature importance scores and ability to capture non-linear relationships. The 80/20 train/test split was stratified on the treatment target. Five-fold cross-validation was run to confirm that the results were stable.

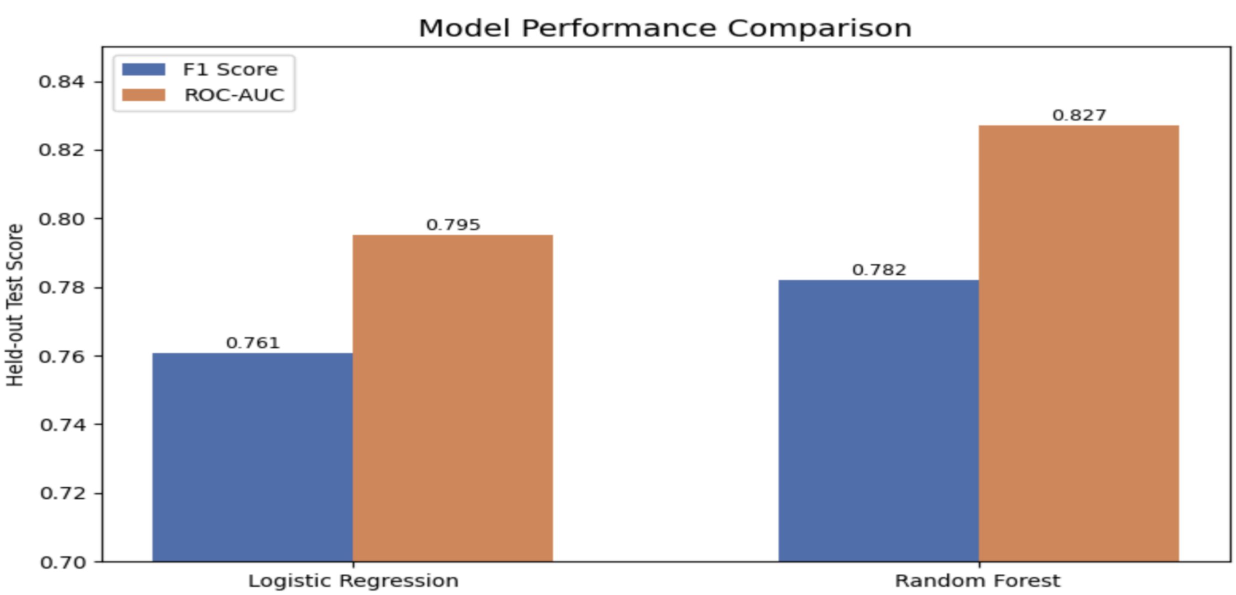
Continuous features were scaled with MinMaxScaler after splitting. The scaler was fit only on the training data and then applied to the test data. This is important because fitting the scaler to the full dataset beforehand would allow test-set information to influence the training process.

**4.2 Model Performance**

| Model | Accuracy | F1 Score | AUC | CV AUC |
|---|---|---|---|---|
| Logistic Regression | 0.738 | 0.761 | 0.795 | 0.791 |
| Random Forest | 0.759 | 0.782 | 0.827 | 0.810 |

Both models perform well above random. The Random Forest outperforms Logistic Regression on every metric. The gap is most visible in AUC, which makes sense because Random Forest captures non-linear relationships that a linear model cannot.

The most important check here is whether cross-validation AUC is close to the test AUC. For both models, these are close, which tells us the models are not overfitting to the train/test split. If the test AUC were much higher than the CV AUC, we would be concerned. A false negative here means predicting someone did not seek treatment when they did. In a mental health context, a false negative is more harmful as they are at risk of going unnoticed.



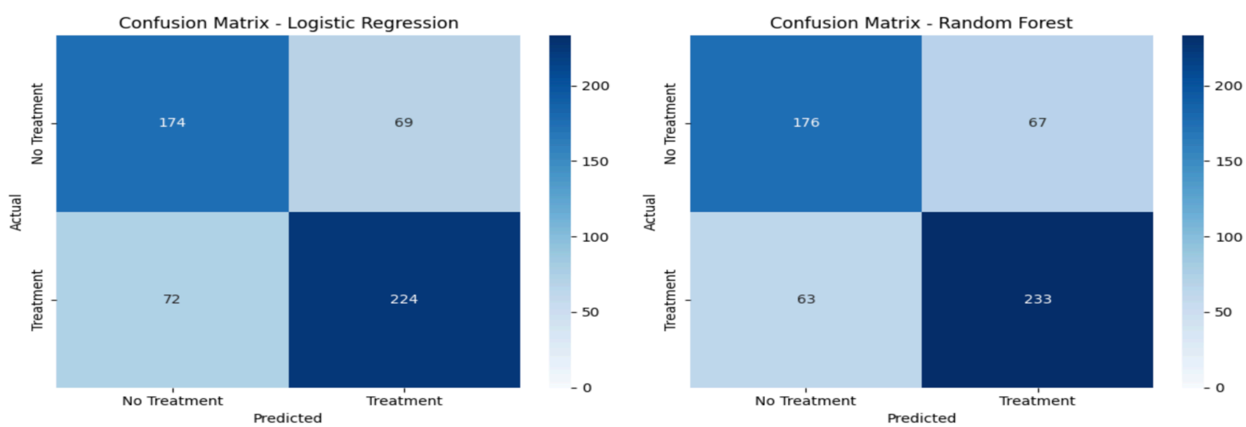
Logistic Regression Classification Report

| Class | Precision | Recall | F1 | Support |
|---|---|---|---|---|
| No Treatment | 0.71 | 0.72 | 0.71 | 243 |
| Treatment | 0.76 | 0.76 | 0.76 | 296 |
| Weighted Average | 0.74 | 0.74 | 0.74 | 539 |

The Logistic Regression classifies treatment seekers slightly better than non-seekers, which is expected given the class imbalance. Precision of 0.76 for treatment means that when the model predicts someone sought treatment, it is correct 76 percent of the time.

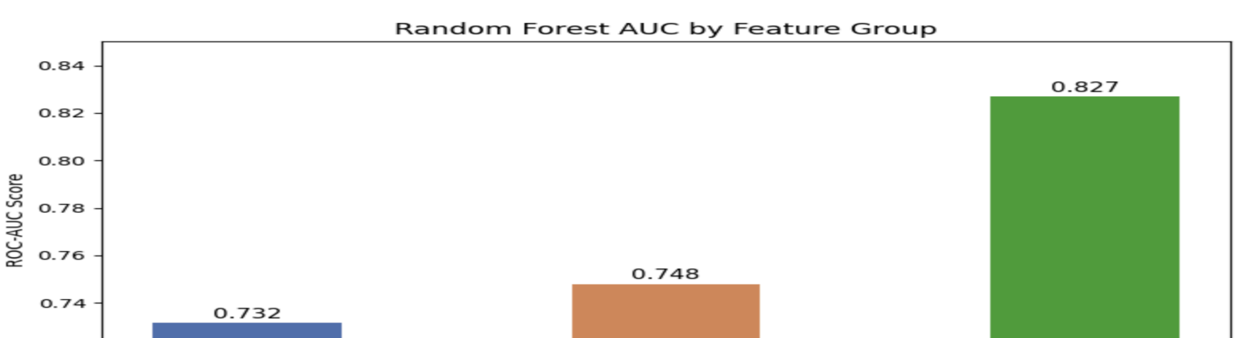
Random Forest Classification Report

| Class | Precision | Recall | F1 | Support |
|---|---|---|---|---|
| No Treatment | 0.74 | 0.72 | 0.73 | 243 |
| Treatment | 0.78 | 0.79 | 0.78 | 296 |
| Weighted Average | 0.76 | 0.76 | 0.76 | 539 |




The Random Forest classifies treatment seekers slightly better than non-seekers, which is expected given the class imbalance. Precision of 0.78 for treatment means that when the model predicts someone sought treatment, it is correct 78 percent of the time.

**4.3 Sub-experiment 1: Personal vs Workplace features**



We wanted to test whether workplace factors or personal factors better predict treatment-seeking. To do this, we trained two separate Random Forest models, one using only personal features and one using only workplace features. Workplace features beat personal features, which confirmed our hypothesis. However, combining both groups improves performance to 0.827, which shows that personal and workplace factors capture different information and both are needed for a complete picture.

**4.4 Sub-experiment 2: Predicting Work Interference Study**
We tried predicting how severely mental health interferes with work as a 4-class problem, which includes Never, Rarely, Sometimes, and Often.

| Class | F1 | Support |
|---|---|---|
| Never | 0.48 | 45 |
| Often | 0.61 | 138 |
| Rarely | 0.07 | 45 |
| Sometimes | 0.57 | 168 |

The model hit 54.3% accuracy against a 25% random baseline, so it is better than guessing. Often and Sometimes are predicted reasonably. They are the most common and have clearer patterns. Rarely is basically unpredictable at F1 of 0.07 because it sits between Never and Sometimes, and the model kept misclassifying it into one of those two. This answers the second part of our research question, which is that we can predict extreme interference levels reasonably well, but the middle ground is hard to distinguish. The features in our dataset do not capture whatever distinguishes someone with rare interference from someone with none or moderate interference, which suggests this boundary is too vague.

**4.5 Sub-experiment 3: Does Survey Year Matter?**
Survey year ranked 8th out of 23 features with an importance score of 0.023. It is not a dominant predictor, but it is not negligible either. Only 7 features ranked above it. This confirms that something measurably changed between 2014 and 2016 in treatment-seeking behavior beyond what the other features explain, which connects directly back to Q1, where treatment rate jumped from 50.6% to 58.5%.

## 5. Visualizations

Vis 1 — Treatment-Seeking Rate by Country
This horizontal bar chart shows the top 15 countries by respondent count, sorted by treatment rate and color-coded by range. The US, Australia, and Switzerland are clearly the highest, while Italy, Russia, and France are the lowest. It supports the geographic variation finding from Q3 and shows that cultural and healthcare context play a significant role, independent of employer behavior.

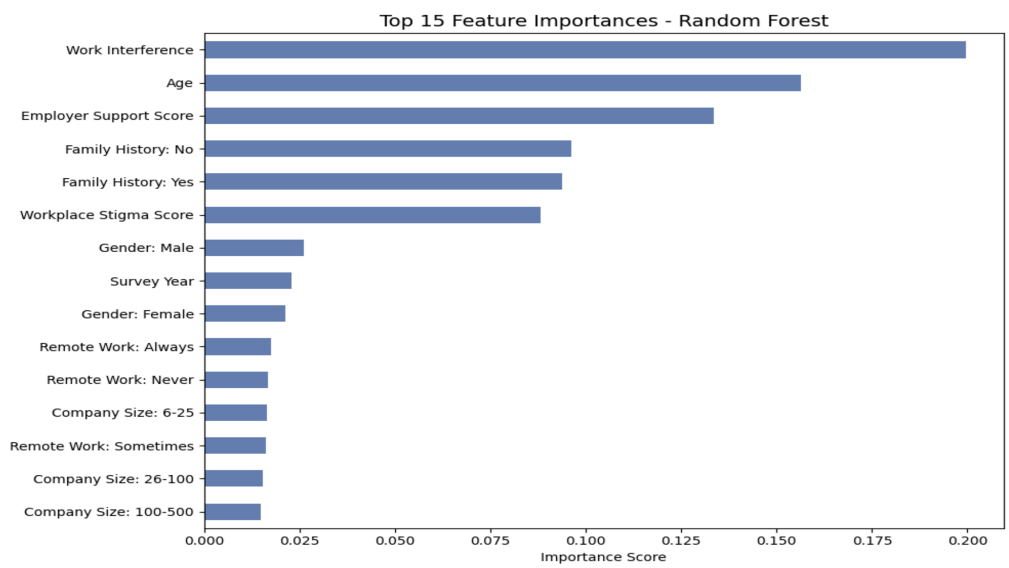
Vis 2 — Year-over-Year Comparison
Three side-by-side bar charts compare 2014 and 2016 across treatment rate, employer support score, and workplace stigma score. Treatment rate increased from 50.6% to 58.5%. This directly answers a part of our research question and shows the two years are genuinely different, justifying survey_year as a model feature.

Vis 3 — Treatment Rate by Employer Support Score Bin
This chart splits respondents into five equal ranges by employer support score and shows the treatment rate within each bin with a trend. The result is a clear upward trend. This is one of the strongest visualizations in the project because it shows a clean relationship between the composite score we engineered and the outcome we are predicting.

Vis 4 — Stigma Score vs Treatment Status
This grouped bar chart divides respondents into Low, Medium, and High Stigma groups and shows the proportion who did and did not seek treatment within each. Higher stigma environments show a higher proportion of non-treatment, directly supporting the stigma_score feature's rank 6 position in the importance chart.

Vis 5 — Willingness to Discuss Mental Health by Relationship Type
This stacked bar chart shows how open people are with coworkers, supervisors, and friends and family. People are most open with friends and family, and the opposite with coworkers. Even in workplace relationships, a large proportion fall into Mixed or Closed categories, which provides important context for why workplace stigma is a meaningful predictor.

Vis 6 — Top 15 Feature Importances
This horizontal bar chart shows the 15 most important features from the Random Forest sorted by importance. Work interference is visually dominant at the top. Both composite scores appear in the top 6.




Vis 7 — Model Performance Comparison
This grouped bar chart compares F1 Score and AUC for both models side by side. Random Forest leads on both. The chart shows that the performance gap is present but not enormous, consistent with both models learning from the same data.

Vis 8 — Confusion Matrices
Two side-by-side confusion matrices show how each model performs at the class level for direct comparison. Both models have similar error profiles with slightly more false negatives on the treatment class. The Random Forest makes fewer errors overall.

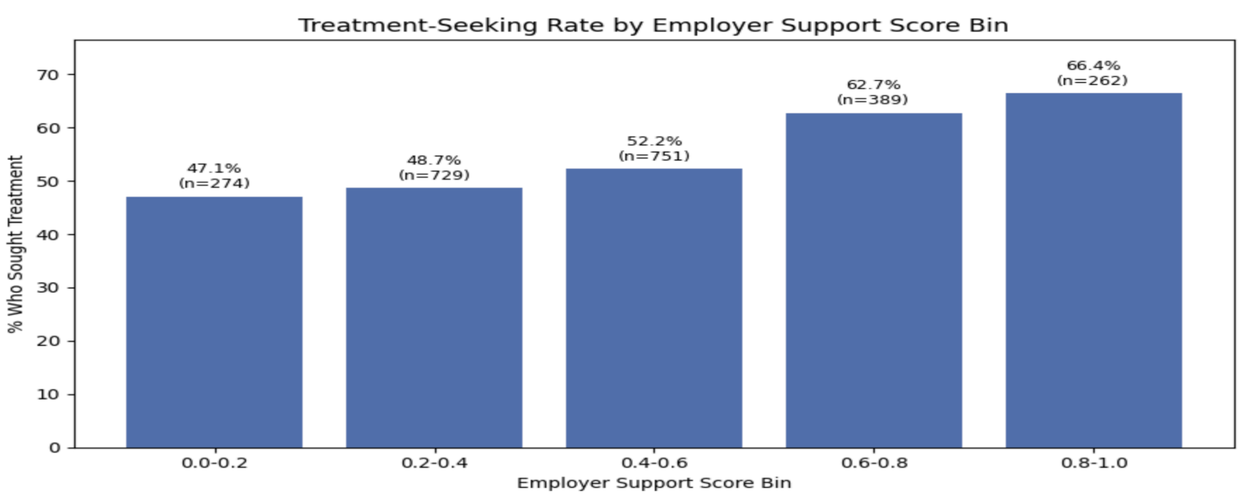
Vis 9 — Personal vs Workplace AUC
This bar chart shows AUC for the personal-only model, workplace-only model, and full model. The visual gap between the two partial models and the full model is the key takeaway, which is that neither group alone is near the full model's performance.

Vis 10 — Age Distribution by Treatment Status
This boxplot shows the age distribution for those who did and did not seek treatment. Medians are close, but treatment-seekers skew slightly older on average. This supports Age being the second most important feature. The model finds age predictive because it separates the two classes across many decision boundaries in the Random Forest.

## 6. Connecting the Findings
The results across the SQL queries, model performances, feature importance, and visualizations help us claim that treatment-seeking in the tech industry is related to how severely mental health is already affecting an individual, how supportive their environment is, and personal factors such as age. Workplace factors outperform personal factors in predicting, but both are needed for an accurate prediction. Across all analysis, the consistent result is that treatment-seeking depends on various factors, employer environment is controllable and predictive, and personal and workplace factors capture information that neither group reveals alone.

## 7. What This Means
Because treatment-seeking appears to be reactive, people who experience frequent mental health interferences are the ones most likely to seek help, which means intervention before the interference becomes a severe issue. Programs that could help workers identify early signs of interference, rather than waiting until it is already affecting their work significantly, could help prevent major issues.

The employer support score ranking being third overall suggests that concrete employer policies matter more for treatment seeking than culture alone. Protections, benefits coverage, wellness programs, and leave policies all contribute to the score, and each is something a company can target to prevent issues.




## 8. Limitations
All data in this dataset is self-reported, meaning that people answered on their own terms. This introduces self-report bias, where respondents may underreport behaviors like avoiding treatment or may overestimate employer support. The true numbers could look different than what is given.

The sampling itself is also biased. OSMI surveys are distributed through mental health networks and tech platforms that are already active on this topic. Someone who has never thought about mental health but is still suffering from it is still unlikely to take the survey. This likely inflates the treatment rates and makes the pattern appear cleaner than it should be in reality.

Geographically, most of the respondents are from the United States, and the survey is in English. The patterns we found are most valid for tech workers in the US, and comparisons with countries that have few respondents should not be taken literally.

Both datasets are from 2014 and 2016, before COVID-19 changed remote work norms and caused a rise in mental health conversations. What we found is a historical baseline, but it could also be a decent reflection of today's time, as mental health is still a serious issue, and its numbers have probably gone up. We acknowledge it is not a representation of the current time, but based on the availability and data, we chose to still go forward with the 2014 and 2016 surveys.

We cannot establish causation. We can say higher employer support is associated with higher treatment rates, but we cannot say that improving employer support causes more people to seek help.

Talking from our coding approach, the biggest issue is how we handled self-employed respondents. In 2016, self-employed people were skipped past all employer-related questions, leaving their workplace columns as NaN. We filled those with 0, which effectively treats them as having zero employer support. This is not accurate since self-employed people have a different workplace support, not an absent one. A better approach could have been to model self-employed respondents separately, but that was outside the general scope.

## 9. Future Work

The most valuable next step would be to incorporate more recent OSMI data. The surveys after 2016 dropped several columns needed for our project. More importantly, post-COVID data would show how remote work and broader mental health awareness shifted patterns. A follow-up study tracking people over time would be stronger as it would allow us to tell if someone's treatment-seeking changed after their current employer improved mental health policies. Tracking individuals across multiple years would give us a better understanding rather than a current snapshot. The comments column we dropped had an 87% null rate. However, if we did run an analysis on them, we could have revealed reasons and patterns that structured queries could not capture. Lastly, the employer support score currently treats all six columns as equally important. A smarter and more effective version would be to weight each column based on its correlation with treatment-seeking.

## 10. Conclusion

This project built a complete data pipeline from two different structurally raw survey datasets to train predictive models, producing findings that answer the main parts of the research question. The most important factors predicting treatment-seeking are how severely mental health already interferes with someone's work, age, employer support score, and family history. Workplace factors are more important than personal factors as predictors, but combining both groups results in more accurate predictions, meaning that individually, both groups do not tell the full story.

The composite scores we engineered, which were employer support and workplace stigma, both rank in the top 6 features and show up consistently in the SQL analysis and the model results. This validates the importance of the features we created, and shows they provide real insight.

The main takeaway is that the tech companies have real leverage over whether their employees seek treatment. Workplace environment is a stronger predictor than personal factors, and the specific policies that make up the employer support score are all things companies can directly control and improve.

Individual Contributions
Vanij and Sean worked closely together throughout the project. Vanij worked on the data pipeline, schema alignment, SCHEMA_MAP design, and model training for both the Logistic Regression and Random Forest. Sean worked on the data cleaning decisions, feature engineering, composite score design, and the three sub-experiments. Visualizations and the report were a shared effort, with both of us building off each other's work and reviewing sections before finalizing. Any major decision was discussed together before being implemented.

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